



Software Quality Assurance Analysts and Testers

15-1253.00

Bright Outlook

Updated 2026

Develop and execute software tests to identify software problems and their causes. Test system modifications to prepare for implementation. Document software and application defects using a bug tracking system and report defects to software or web developers. Create and maintain databases of known defects. May participate in software design reviews to provide input on functional requirements, operational characteristics, product designs, and schedules.

Sample of reported job titles: Automation Tester, Information Technology Analyst (IT Analyst), Quality Assurance Analyst (QA Analyst), Quality Assurance Engineer (QA Engineer), Quality Engineer, Software Quality Assurance Analyst (SQA Analyst), Software Quality Assurance Engineer (SQA Engineer), Software Quality Engineer, Software Test Engineer, Test Engineer

- Summary
- Details
- Custom
- Easy Read
- Veterans
- Español

Occupation-Specific Information

Tasks

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Importance	Category	Task
93	Core	Identify, analyze, and document problems with program function, output, online screen, or content.
92	Core	Document software defects, using a bug tracking system, and report defects to software developers.
88	Core	Develop testing programs that address areas such as database impacts, software scenarios, regression testing, negative testing, error or bug retests, or usability.
84	Core	Design test plans, scenarios, scripts, or procedures.

Importance	Category	Task
83	Core	+ Document test procedures to ensure replicability and compliance with standards.
82	Core	+ Provide feedback and recommendations to developers on software usability and functionality.
82	Core	+ Install, maintain, or use software testing programs.
80	Core	+ Test system modifications to prepare for implementation.
79	Core	+ Create or maintain databases of known test defects.
78	Core	+ Develop or specify standards, methods, or procedures to determine product quality or release readiness.

Occupational Requirements

Work Activities

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Importance	Work Activity
99	+ Working with Computers — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
91	+ Getting Information — Observing, receiving, and otherwise obtaining information from all relevant sources.
86	+ Communicating with Supervisors, Peers, or Subordinates — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.
85	+ Processing Information — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
84	+ Updating and Using Relevant Knowledge — Keeping up-to-date technically and applying new knowledge to your job.
82	+ Analyzing Data or Information — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.

Contents

75% Every day

22% Once a week or more but not every day

+ **Spend Time Sitting** — How much does this job require sitting?

71% Continually or almost continually

26% More than half the time

+ **Importance of Being Exact or Accurate** — How important is being very exact or highly accurate in performing this job?

50% Extremely important

44% Very important

+ **Work With or Contribute to a Work Group or Team** — How important is it to work with or contribute to a work group or team in this job?

58% Extremely important

27% Very important

15% Important

+ **Freedom to Make Decisions** — How much decision making freedom, without supervision, does the job offer?

24% A lot of freedom

65% Some freedom

12% Limited freedom

+ **Indoors, Environmentally Controlled** — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?

67% Every day

12% Once a week or more but not every day

15% Never

+ **Telephone Conversations** — How often do you have telephone conversations in this job?

42% Every day

42% Once a week or more but not every day

+ **Contact With Others** — How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?

29% Constant contact with others

47% Contact with others most of the time

18% Contact with others about half the time

+ **Determine Tasks, Priorities and Goals** — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?

12% A lot of freedom

74% Some freedom

Experience Requirements

Job Zone

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Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, conservation scientists, art directors, and cost estimators.
SVP Range	(7.0 to < 8.0)

Training & Credentials

State training	Select a State Go
Local training	ZIP Code Go
Certifications	Find certifications nationwide
State licenses	Select a State Go

Apprenticeship Opportunities

 All 1 displayed

Example apprenticeship titles for this occupation:

- Software Quality Assurance Tester (Nof)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit [Apprenticeship.gov](https://www.apprenticeship.gov) to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- + **Application server software** — Atlassian Bitbucket ; GitLab ; Red Hat OpenShift ; Spring Boot
- + **Configuration management software** — Chef ; Perforce Helix software; Puppet ; Visible Razor
- + **Data base management system software** — Amazon DynamoDB ; Elasticsearch ; MongoDB ; Oracle PL/SQL
- + **Data base user interface and query software** — Airtable; Apache Hive ; Blackboard software; IBM DB2
- + **Development environment software** — Apache Kafka ; Apache Maven ; Apache Subversion SVN ; Oracle Java 2 Platform Enterprise Edition J2EE
- + **Enterprise resource planning ERP software** — Microsoft Dynamics; Oracle Fusion Applications; Oracle PeopleSoft ; SAP software
- + **Object or component oriented development software** — Apache Spark ; jQuery ; Scala ; Swift
- + **Operating system software** — Google Android ; Microsoft Windows Server ; Red Hat Enterprise Linux ; UNIX Shell
- + **Program testing software** — Hewlett Packard LoadRunner; IBM Rational Robot; JUnit ; Selenium
- + **Web platform development software** — Django ; Google Angular ; React ; Spring Framework

Hot Technologies are requirements most frequently included across all employer job postings.
[See all 126 Hot Technologies for this occupation.](#)

In Demand skills are frequently included in employer job postings for this occupation.
[See all 24 In Demand skills for this occupation.](#)

Essential Skills

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Importance	Essential Skill
75	+ Reading Comprehension — Understanding written sentences and paragraphs in work-related documents.

Importance	Essential Skill
72	+ Active Listening — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
72	+ Critical Thinking — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
72	+ Speaking — Talking to others to convey information effectively.
69	+ Writing — Communicating effectively in writing as appropriate for the needs of the audience.
66	+ Monitoring — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
53	+ Active Learning — Understanding the implications of new information for both current and future problem-solving and decision-making.
47	+ Learning Strategies — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
44	+ Mathematics — Using mathematics to solve problems.
44	+ Science — Using scientific rules and methods to solve problems.

Transferable Skills

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Importance	Transferable Skill
66	+ Programming — Writing computer programs for various purposes.
60	+ Complex Problem Solving — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
60	+ Quality Control Analysis — Conducting tests and inspections of products, services, or processes to evaluate quality or performance.
56	+ Systems Evaluation — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.

Importance	Transferable Skill
53	+ Judgment and Decision Making — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
53	+ Operations Analysis — Analyzing needs and product requirements to create a design.
53	+ Systems Analysis — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
50	+ Coordination — Adjusting actions in relation to others' actions.
50	+ Operations Monitoring — Watching gauges, dials, or other indicators to make sure a machine is working properly.
50	+ Time Management — Managing one's own time and the time of others.

Knowledge

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Importance	Knowledge
85	+ Computers and Electronics — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
70	+ English Language — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
57	+ Engineering and Technology — Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services.
57	+ Mathematics — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
50	+ Design — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.

Importance	Knowledge
47	Education and Training — Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects.
42	Customer and Personal Service — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
36	Administration and Management — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources.
35	Administrative — Knowledge of administrative and office procedures and systems such as word processing, managing files and records, stenography and transcription, designing forms, and workplace terminology.
30	Telecommunications — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

50%	Bachelor’s degree required
26%	Associate’s degree required
9%	Post-secondary certificate required

Worker Characteristics

Abilities

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Importance	Ability
75	Deductive Reasoning — The ability to apply general rules to specific problems to produce answers that make sense.

Importance	Ability
75	Inductive Reasoning — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
75	Oral Comprehension — The ability to listen to and understand information and ideas presented through spoken words and sentences.
75	Oral Expression — The ability to communicate information and ideas in speaking so others will understand.
75	Problem Sensitivity — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
75	Written Comprehension — The ability to read and understand information and ideas presented in writing.
72	Written Expression — The ability to communicate information and ideas in writing so others will understand.
69	Information Ordering — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
66	Near Vision — The ability to see details at close range (within a few feet of the observer).
56	Speech Recognition — The ability to identify and understand the speech of another person.

Career Interest Types

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Occupational Interest	Career Interest Type
79	Investigative — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.

Occupational Interest	Career Interest Type
78	Conventional — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
47	Realistic — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
12	Artistic — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art.
10	Enterprising — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law.
9	Social — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities.

Specific Interest Areas

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Occupational Interest	Specific Interest Area
92	Information Technology — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
48	Engineering — Work involves applying science and technology to the design, building, testing, and use of electrical and electronic components, mechanical

Occupational Interest	Specific Interest Area
	devices and machines, automotive, marine, and aerospace equipment and vehicles, materials, or structures.
37	Mathematics/Statistics — Work involves developing and applying mathematics and statistical theory and methods.

Work Styles

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Impact	Work Style
100	Attention to Detail — A tendency to be detail-oriented, organized, and thorough in completing work.
80	Dependability — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.
77	Cautiousness — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
68	Intellectual Curiosity — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
59	Achievement Orientation — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards.
59	Integrity — A tendency to be honest and ethical at work.
59	Perseverance — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work.
54	Adaptability — A tendency to be open to and comfortable with change, new experiences, or ideas at work.
43	Innovation — A tendency to be inventive, to be imaginative, and to adopt new perspectives on ways to accomplish work.
40	Cooperation — A tendency to be pleasant, helpful, and willing to assist others at work.

Workforce Characteristics

Wages & Employment Trends

Median wages (2024) \$49.33 hourly, \$102,610 annual

State wages

Select a State

Go

Local wages

ZIP Code

Go

Employment (2024) 201,700 employees

Projected growth (2024-2034) Much faster than average (7% or higher)

Projected job openings (2024-2034) 14,000

State trends

Select a State

Go

Top industries (2024) [Professional, Scientific, and Technical Services](#) (41% employed in this sector)
[Information](#) (19%)
[Finance and Insurance](#) (12%)

Source: Bureau of Labor Statistics [2024 wage data](#) and [2024-2034 employment projections](#). “Projected growth” represents the estimated change in total employment over the projections period (2024-2034). “Projected job openings” represent openings due to growth and replacement.

Job Openings on the Web

State job openings

Select a State

Go

Local job openings

ZIP Code

Go

More Information

Related Occupations

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- 15-1299.07 [Blockchain Engineers](#) **Bright Outlook**
- 17-2061.00 [Computer Hardware Engineers](#)
- 15-1251.00 [Computer Programmers](#)

- 15-1211.00 [Computer Systems Analysts](#) ☀
- 15-1299.08 [Computer Systems Engineers/Architects](#) ☀
- 15-1243.00 [Database Architects](#) ☀
- 15-1299.05 [Information Security Engineers](#) ☀
- 15-1299.04 [Penetration Testers](#) ☀
- 15-1252.00 [Software Developers](#) ☀
- 17-2112.02 [Validation Engineers](#) ☀

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

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- [American Society for Quality](#) ↗
- [Association for Computing Machinery](#) ↗
- [Association of Software Testing](#) ↗
- [Computing Research Association](#) ↗
- [IEEE Computer Society](#) ↗

Accreditation, Certification, & Unions

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- [American Software Testing Qualifications Board](#) ↗ **ALLY**
- [Association for Testing and Software Quality Assurance](#) ↗ **ALLY**
- [CompTIA](#) ↗
- [Institute for Certification of Computing Professionals](#) ↗
- [International Software Testing Qualifications Board](#) ↗